Updated schedule

Week 7: May 26-27	Event-Based Semantics (Lasersohn, 2012)	Event Semantics	Exercise 4: Generalized Quantifiers
Week 8: June 2-3	None	Lexical Semantics	Exercise 5: Event Semantics
Week 9: June 9-10	Dynamic Semantics (Stanford Encyclopedia of Philosophy)	Dynamic Semantics	Exercise 6: Lexical Semantics
Week 10: June 16-17	Discourse Representation Theory (Stanford Encyclopedia of Philosophy)	DRT	Exercise 7: Dynamic Semantics
Week 11: June 23-24	None	Presuppositions in DRT	Exercise 8: DRT
Week 12: June 30 - July 1	None	Current Issues and Applications	Exercise 9: Presuppositions in DRT
Week 13: July 7-8	None	Exam Review	Take-home Practice Exam
Week 14: July 14-15	None	**Exam**	No lecture

(old)

Week 7: May 26-27	Event-Based Semantics (Lasersohn, 2012)	Event Semantics	Exercise 4: Generalized Quantifiers
Week 8: June 2-3	None	Lexical Semantics	Exercise 5: Event Semantics
Week 9: June 9-10	Dynamic Semantics (Stanford Encyclopedia of Philosophy)	Dynamic Semantics	Exercise 6: Lexical Semantics
Week 10: June 16-17	Discourse Representation Theory (Stanford Encyclopedia of Philosophy)	DRT	Exercise 7: Dynamic Semantics
Week 11: June 23-24	None	Presuppositions in DRT; Take-home Practice Exam Assigned	Exercise 8: DRT
Week 12: June 30 - July 1	None	Current Issues and Applications; Exam Review	Exercise 9: Presuppositions in DRT; Take-home Practice Exam
Week 13: July 7-8	None	No lecture	No lecture
Week 14: July 14-15	None	Exam Review	**Exam**

(new)

https://mjs227.github.io/courses/semantic-theory-26/schedule

Updated schedule

- Take-home exam:
  - assigned ~~June 30~~ June 23
  - discussion ~~July 8~~ July 1

- Exam review: ~~July 7-8~~ June 30, July 1, July 14

- Exam: ~~July 14~~ July 15

- **Keep in mind**: I won't be able to hold office hours on weeks 13-14

Dynamic Semantics

Week 8

Slides and materials based on the courses by
Noortje Venhuizen and Mareike Hartmann

A problem* for type theory: context-dependent expressions

- **Deictic** expressions depend on the *physical* utterance situation:
  - *“I”*, *“you”*, *“now”*, *“here”*, *“this”*, …

- **Anaphoric** expressions refer to the *linguistic context*/previous discourse:
  - *“he”*, *“she”*, *“it”*, *“then”*, …

More context-dependent expressions

- Context dependence is a pervasive property of natural language:
  - *“every student must be familiar with the basic properties of FOL”*
  - *“it is rainy everywhere”*
  - *“John is always late”*
  - *“Bill bought an expensive car”*
  - *“another one, please!”*
  - *“the student is working”*

Context theory

- Natural-language expressions can vary their meaning with context: *“I”*, *“you”*, *“here”*, *“this”*, *“now”*, etc.

- A simple context theory (based on Lewis 1970/1972):
  - Model contexts as “vectors”: sequences of semantically relevant context data with fixed arity
  - Model meanings as functions from contexts to denotations—more specifically, as functions from specific context components to denotations

Defining a context "vector"

- **Context** c = (a, b, ℓ, t, r)
  - a = speaker
  - b = addressee
  - ℓ = utterance location
  - t = utterance time
  - r = referred object

$$[\![\mathit{I}]\!]^{M,g,c} = \mathrm{utt}(c) = a$$

$$[\![\mathit{you}]\!]^{M,g,c} = \mathrm{adr}(c) = b$$

$$[\![\mathit{here}]\!]^{M,g,c} = \mathrm{loc}(c) = l$$

$$[\![\mathit{now}]\!]^{M,g,c} = \mathrm{time}(c) = t$$

$$[\![\mathit{this}]\!]^{M,g,c} = \mathrm{ref}(c) = r$$

Another problem for traditional type theory: donkeys
(Compositional derivation of “donkey sentences”)

- Indefinite noun phrases and conditionals interact strangely, e.g.:

“if a farmer owns a donkey, he beats it”

Another problem for traditional type theory: donkeys

(Compositional derivation of "donkey sentences")

- Indefinite noun phrases and conditionals interact strangely, e.g.:

  "*if a farmer owns a donkey, he beats it*"

- ~~$\exists x \exists y[\mathit{farmer}(x) \land \mathit{donkey}(y) \land \mathit{own}(x, y)] \rightarrow \mathit{beat}(x, y)$~~
  - not closed (x and y occur free)

- ~~$\exists x \exists y[(\mathit{farmer}(x) \land \mathit{donkey}(y) \land \mathit{own}(x, y)) \rightarrow \mathit{beat}(x, y)]$~~
  - wrong truth conditions (much too weak)

- $\forall x \forall y[(\mathit{farmer}(x) \land \mathit{donkey}(y) \land \mathit{own}(x, y)) \rightarrow \mathit{beat}(x, y)]$
  - Correct! But how can it be derived compositionally?

Anaphora and compositionality

- Consider the following sentence: S_1 = "*a person who works hard is happy*"
    - we can interpret S_1 as*: $\exists x[\mathit{person}(x) \land \mathit{work\text{-}hard}(x) \land \mathit{happy}(x)]$

- Now consider S_2 = "*<u>a person</u>$_i$ works hard*", S_3 = "*they$_i$ are happy*"

- $S_2 S_3 = S_1$, but can we derive this compositionally?
    - $\exists x[\mathit{person}(x) \land \mathit{work\text{-}hard}(x)] \land \mathit{happy}(\color{red}{x})$

Yet another problem: what are indefinites?

- Option I: Existential quantifiers? (cf. Russell, 1919)
    - No: donkey sentences

- Option II: Universal quantifiers?
    - No: *"I saw <u>a</u> donkey"* $\nvDash$ *"I saw <u>every</u> donkey"*

- Option III: Ambiguous?

We need to re-think our logic

- We need a new way to think about quantification and free variables: ***Dynamic* Predicate Logic** (DPL; Groenendijk and Stokhof, 1991)

DPL: Syntax

- **Terms**: $TERM = VAR \cup CON$
- **Atomic formulas**:
  - $R(t_1, \ldots, t_n)$ for $R \in PRED^n$ and $t_1, \ldots, t_n \in TERM$

- **Well-formed formulas** (WFF):
  - All atomic formulas are WFFs
  - If $x \in VAR$, then $\exists x$ is a WFF
    - *Note*: there is *no* quantifier scope in DPL—*all* variables are free
  - If φ and ψ are WFFs, then $\sim\varphi$ and $(\varphi \cdot \psi)$ are WFFs
  - Nothing else is a WFF

(Rough) correspondences with FOL

- $\neg\varphi \approx {\sim}\varphi$
- $\varphi \wedge \psi \approx \varphi \cdot \psi$
- $\varphi \vee \psi \approx {\sim}({\sim}\varphi \cdot {\sim}\psi)$
- $\varphi \rightarrow \psi \approx {\sim}(\varphi \cdot {\sim}\psi)$
- $\exists x\varphi \approx \exists x \cdot \varphi$
- $\forall x\varphi \approx \exists x \rightarrow \varphi = {\sim}(\exists x \cdot {\sim}\varphi)$

DPL: examples

- "*A person who works hard is happy.*":
  - $\exists x \cdot \textit{person}(x) \cdot \textit{work-hard}(x) \cdot \textit{happy}(x)$

- "*A person works hard. They are happy.*":
  - $(\exists x \cdot \textit{person}(x) \cdot \textit{work-hard}(x)) \cdot (\textit{happy}(x))$
    $= \exists x \cdot \textit{person}(x) \cdot \textit{work-hard}(x) \cdot \textit{happy}(x)$

- "*If a farmer owns a donkey, he beats it.*":
  - $(\exists x \cdot \textit{farmer}(x) \cdot \exists y \cdot \textit{donkey}(y) \cdot \textit{own}(x, y)) \rightarrow \textit{beat}(x, y)$
    $= {\sim}((\exists x \cdot \textit{farmer}(x) \cdot \exists y \cdot \textit{donkey}(y) \cdot \textit{own}(x, y)) \cdot {\sim}\textit{beat}(x, y))$

DPL: semantics (one way to do it)

- We can *map DPL formulas into FOL* for interpretation. For a DPL formula φ, define its corresponding FOL formula φ° as $\langle\varphi\rangle\top$, where:
  1. $\langle\bot\rangle\psi = \bot$
  2. $\langle\top\rangle\psi = \top$
  3. $\langle P(x_1, \ldots, x_n)\rangle\psi = P(x_1, \ldots, x_n) \wedge \psi$
  4. $\langle\exists x\rangle\psi = \exists x[\psi]$
  5. $\langle\varphi_1 \cdot \varphi_2\rangle\psi = \langle\varphi_1\rangle(\langle\varphi_2\rangle\psi)$
  6. $\langle{\sim}\varphi\rangle\psi = \neg(\langle\varphi\rangle\top) \wedge \psi$

(Rough) correspondences with FOL (again)

- $(\sim\varphi)^\circ = \langle\sim\varphi\rangle\top = \neg(\langle\varphi\rangle\top) \wedge \top = \neg(\varphi \wedge \top) \wedge \top = \neg\varphi$
- $(\varphi \cdot \psi)^\circ = \langle\varphi \cdot \psi\rangle\top = \langle\varphi\rangle(\langle\psi\rangle\top) = \langle\varphi\rangle(\psi \wedge \top) = \varphi \wedge \psi \wedge \top = \varphi \wedge \psi$
- $(\sim(\sim\varphi \cdot \sim\psi))^\circ = \neg(\neg\varphi \wedge \neg\psi) = \varphi \vee \psi$
- $(\varphi \rightarrow \psi)^\circ = (\sim(\varphi \cdot \sim\psi))^\circ = \neg(\varphi \wedge \neg\psi) = \varphi \rightarrow \psi$
- $(\exists x \cdot \varphi)^\circ = \langle\exists x\rangle(\langle\varphi\rangle\top) = \langle\exists x\rangle(\varphi \wedge \top) = \exists x[\varphi \wedge \top] = \exists x[\varphi]$
- $(\exists x \rightarrow \varphi)^\circ = (\sim(\exists x \cdot \sim\varphi))^\circ = \neg(\langle\exists x \cdot \sim\varphi\rangle\top) \wedge \top = \neg(\langle\exists x\rangle(\neg\varphi \wedge \top \wedge \top)) \wedge \top = \neg\exists x[\neg\varphi \wedge \top \wedge \top] \wedge \top = \neg\exists x[\neg\varphi] = \forall x[\varphi]$

DPL semantics: example

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$(\exists x \cdot P(x) \cdot W(x) \cdot H(x))^\circ$

$= \langle \exists x \cdot P(x) \cdot W(x) \cdot H(x) \rangle \top$

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$((\exists x \cdot F(x) \cdot \exists y \cdot D(y) \cdot O(x, y)) \rightarrow B(x, y))^{\circ}$

$= ({\sim}((\exists x \cdot F(x) \cdot \exists y \cdot D(y) \cdot O(x, y)) \cdot {\sim}B(x, y)))^{\circ}$

$= \langle {\sim}((\exists x \cdot F(x) \cdot \exists y \cdot D(y) \cdot O(x, y)) \cdot {\sim}B(x, y)) \rangle \top$

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5. $\langle \varphi_1 \cdot \varphi_2 \rangle \psi = \langle \varphi_1 \rangle (\langle \varphi_2 \rangle \psi)$
6. $\langle {\sim}\varphi \rangle \psi = \neg(\langle \varphi \rangle \top) \wedge \psi$

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DPL semantics: another example

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$((\exists x \cdot F(x) \cdot \exists y \cdot D(y) \cdot O(x, y)) \rightarrow B(x, y))^\circ$

$= \neg \exists x[F(x) \land \exists y[D(y) \land O(x, y) \land \neg B(x, y)]]$

$\leftrightarrow \forall x[F(x) \rightarrow \forall y[(D(y) \land O(x, y)) \rightarrow B(x, y)]$

Meanwhile at the philosophy department…

- Asking the big questions: what is meaning?
    - Truth-conditions vs. context-change
    - Sentence vs. discourse
    - Semantics vs. pragmatics

A new perspective on meaning

- Truth-conditional Semantics ⇒ ***Dynamic* Semantics**

- Basic semantic value:
  - truth-conditions ⇒ context-change potential
- (In)definite NPs are:
  - quantificational ⇒ variables
- Existential quantification over:
  - A sentence ⇒ the discourse
- Quantification is:
  - selective ⇒ unselective

Meaning as context-change potential

- In dynamic semantics, the meaning of an expression is the effect it has on its context: context ⇔ meaning
  - context changes meaning
  - meaning changes context

- Note: This is a *generalisation of* rather than an *alternative to* classical truth-conditional semantics

Discourse variables and quantification

- Division of labor between definite and indefinite NPs:
  - Indefinite NPs introduce discourse referents, which can serve as antecedents for anaphoric reference
  - Definite NPs refer to “old” or “familiar” discourse referents (which are already part of the meaning representation)

- E.g.: *“A dog came in. It barked.”* $\mapsto$ *dog*(x) $\wedge$ *came-in*(x) $\wedge$ *barked*(x)
  - … true iff there is a value for x that verifies the conditions

Unselective quantification

"*every* *farmer who owns a donkey* *beats it*"

(quantifier) (restriction) (nuclear scope)

- This is true iff for every value assignment to x and y:
  - if $[\![\mathit{farmer}(x) \wedge \mathit{donkey}(y) \wedge \mathit{owns}(x, y)]\!]^{M,g} = 1$ then $[\![\mathit{beats}(x, y)]\!]^{M,g} = 1$
- Quantification is restricted to those individuals who satisfy the restriction
  - Quantification is ***unselective***, i.e. all free variables are bound.